

Module 3 - Syntax Parsing

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- Tagsets for English
- Part of speech tagging
- Rule based part-of-speech tagging
- Stochastic part-of speech tagging
- Transformation Based tagging

Tagsets for English

- ▶ There are a small number of popular tagsets for English
- ▶ The choice of Tagsets depends on the nature of the application
 - ▶ Small Tagsets (more general)
 - ▶ Large Tagsets (finer tags)

Tagsets for English

- ▶ Some of the widely used Part of Speech Tagsets are
 - ▶ 45-tag Penn Treebank tagset
 - ▶ 87-tag tagset used for the Brown corpus
 - ▶ medium-sized 61 tag C5 tagset
 - ▶ 146-tag C7 tagset
- ▶ Each word is associated with the tag from the used Tagsets.

Tag	Description	Example	Tag	Description	Example
CC	Coordin. Conjunction	<i>and, but, or</i>	SYM	Symbol	<i>+, %, &</i>
CD	Cardinal number	<i>one, two, three</i>	TO	“to”	<i>to</i>
DT	Determiner	<i>a, the</i>	UH	Interjection	<i>ah, oops</i>
EX	Existential ‘there’	<i>there</i>	VB	Verb, base form	<i>eat</i>
FW	Foreign word	<i>mea culpa</i>	VBD	Verb, past tense	<i>ate</i>
IN	Preposition/sub-conj	<i>of, in, by</i>	VBG	Verb, gerund	<i>eating</i>
JJ	Adjective	<i>yellow</i>	VBN	Verb, past participle	<i>eaten</i>
JJR	Adj., comparative	<i>bigger</i>	VBP	Verb, non-3sg pres	<i>eat</i>
JJS	Adj., superlative	<i>wildest</i>	VBZ	Verb, 3sg pres	<i>eats</i>
LS	List item marker	<i>1, 2, One</i>	WDT	Wh-determiner	<i>which, that</i>
MD	Modal	<i>can, should</i>	WP	Wh-pronoun	<i>what, who</i>
NN	Noun, sing. or mass	<i>llama</i>	WP\$	Possessive wh-	<i>whose</i>
NNS	Noun, plural	<i>llamas</i>	WRB	Wh-adverb	<i>how, where</i>
NNP	Proper noun, singular	<i>IBM</i>	\$	Dollar sign	<i>\$</i>
NNPS	Proper noun, plural	<i>Carolinas</i>	#	Pound sign	<i>#</i>
PDT	Predeterminer	<i>all, both</i>	“	Left quote	<i>(‘ or “</i>
POS	Possessive ending	<i>’s</i>	”	Right quote	<i>(’ or ”</i>
PP	Personal pronoun	<i>I, you, he</i>	(	Left parenthesis	<i>([, ({ , <</i>
PP\$	Possessive pronoun	<i>your, one’s</i>	)	Right parenthesis	<i>(] ,) , } , ></i>
RB	Adverb	<i>quickly, never</i>	,	Comma	<i>,</i>
RBR	Adverb, comparative	<i>faster</i>	.	Sentence-final punc	<i>(. ! ?)</i>
RBS	Adverb, superlative	<i>fastest</i>	:	Mid-sentence punc	<i>(: ; ... – -)</i>
RP	Particle	<i>up, off</i>			

Figure 8.6 Penn Treebank Part-of-Speech Tags (Including Punctuation)

Contd...

- ▶ An example of a tagged sentence from the Penn Treebank version of the Brown corpus

The/DT grand/JJ jury/NN commented/VBD on/IN a/DT
number/NN of/IN other/JJ topics/NNS ./.

Part-of-speech tagging

- ▶ Part-of-speech tagging (or just tagging for short) is the process of assigning a part-of-speech or other lexical class marker to each word in a corpus.
- ▶ Tags are also usually applied to punctuation markers; thus tagging for natural language is the same process as tokenization for computer languages, although tags for natural languages are much more ambiguous.

Contd...

Example:

VB DT NN .

Book that flight .

VBZ DT NN VB NN ?

Does that flight serve dinner ?

Contd...

- ▶ Even in these simple examples, automatically assigning a tag to each word is not trivial.
- ▶ For example, **book is ambiguous** - more than one possible usage and part of speech.
- ▶ It can be a verb (as in book that flight or to book the suspect) or a noun (as in hand me that book, or a book of matches).
- ▶ Similarly **that** can be a determiner (as in Does that flight serve dinner), or a complementizer (as in I thought that your flight was earlier).

Contd...

- ▶ Words often have more than one POS: *back*
 - ▶ The *back* door = JJ
 - ▶ On my *back* = NN
 - ▶ Win the voters *back* = RB
 - ▶ Promised to *back* the bill = VB
- ▶ The POS tagging problem is to determine the POS tag for a particular instance of a word.

Contd...

- ▶ The problem of POS-tagging is to resolve these ambiguities, choosing the proper tag for the context.
- ▶ Part-of-speech tagging is thus one of the many disambiguation tasks

How Hard is POS Tagging? Measuring Ambiguity

		87-tag Original Brown	45-tag Treebank Brown
Unambiguous (1 tag)		44,019	38,857
Ambiguous (2–7 tags)		5,490	8844
Details:	2 tags	4,967	6,731
	3 tags	411	1621
	4 tags	91	357
	5 tags	17	90
	6 tags	2 (<i>well, beat</i>)	32
	7 tags	2 (<i>still, down</i>)	6 (<i>well, set, round, open, fit, down</i>)
	8 tags		4 (<i>'s, half, back, a</i>)
	9 tags		3 (<i>that, more, in</i>)

Quick Test: Agreement?

1. the students went to class
2. plays well with others
3. fruit flies like a banana

DT: the, this, that

NN: noun

VB: verb

P: preposition

ADV: adverb

Quick Test

- ▶ the students went to class

DT NN VB P NN

- ▶ plays well with others

VB ADV P NN

NN NN P DT

fruit flies like a banana

NN NN VB DT NN

NN VB P DT NN

NN NN P DT NN

NN VB VB DT NN

Methods for POS Tagging

1. Rule-based tagging – uses hand written rules
 - ▶ ENGTWOL (ENGLISH TWO Level analysis)
2. Stochastic- Probabilistic sequence models
 - ▶ HMM (Hidden Markov Model) tagging
 - ▶ MEMMs (Maximum Entropy Markov Models)
3. Transformation Based Tagging – uses ruled learned automatically.

Rule-based POS tagging

- ▶ First stage - used a dictionary to assign each word a list of potential parts of speech.
- ▶ Second stage - used large lists of hand-written disambiguation rules to winnow down this list to a single part-of-speech for each word.

Rule-based POS tagging

ENGTWOL is a rule tagger

- Uses two level lexicon transducer
- Uses hand crafted rules (about 1 100 rules)

Sample lexical entries from the ENGTWOL lexicon

Word	POS	Additional POS features
smaller	ADJ	COMPARATIVE
entire	ADJ	ABSOLUTE ATTRIBUTIVE
fast	ADV	SUPERLATIVE
that	DET	CENTRAL DEMONSTRATIVE SG
all	DET	PREDETERMINER SG/PL QUANTIFIER
dog's	N	GENITIVE SG
furniture	N	NOMINATIVE SG NOINDEFDETERMINER
one-third	NUM	SG
she	PRON	PERSONAL FEMININE NOMINATIVE SG3
show	V	IMPERATIVE VFIN
show	V	PRESENT -SG3 VFIN
show	N	NOMINATIVE SG
shown	PCP2	SVOO SVO SV
occurred	PCP2	SV
occurred	V	PAST VFIN SV

Contd...

SG - singular

SG3- other than third-person-singular.

ABSOLUTE - non-comparative and non-superlative for an adjective

NOMINATIVE - non-genitive

PCP2 - past participle.

PRE, CENTRAL, and POST - ordering slots for determiners

NOINDEFDETERMINER - words like furniture do not appear with the indefinite determiner

SV, SVO, and SVOO - the subcategorization or complementation pattern for the verb

Contd...

- ▶ For example the phrase *“Pavlov had shown that salivation”*

Pavlov	PAVLOV N NOM SG PROPER
had	HAVE V PAST VFIN SVO HAVE PCP2 SVO
shown	SHOW PCP2 SVOO SVO SV
that	ADV PRON DEM SG DET CENTRAL DEM SG CS
salivation	N NOM SG
...	

Contd...

- ▶ A set of about 1,100 constraints are then applied to the input sentence to rule out incorrect parts of speech.
- ▶ For example one constraint eliminates all readings of that except the ADV (adverbial intensifier) sense (this is the sense in the sentence it isn't that odd).

A simplified version of the constraint

ADVERBIAL-THAT RULE

Given input: "that"

if

(+1 A/ADV/QUANT); / * *if next word is adj, adverb, or quantifier* * /

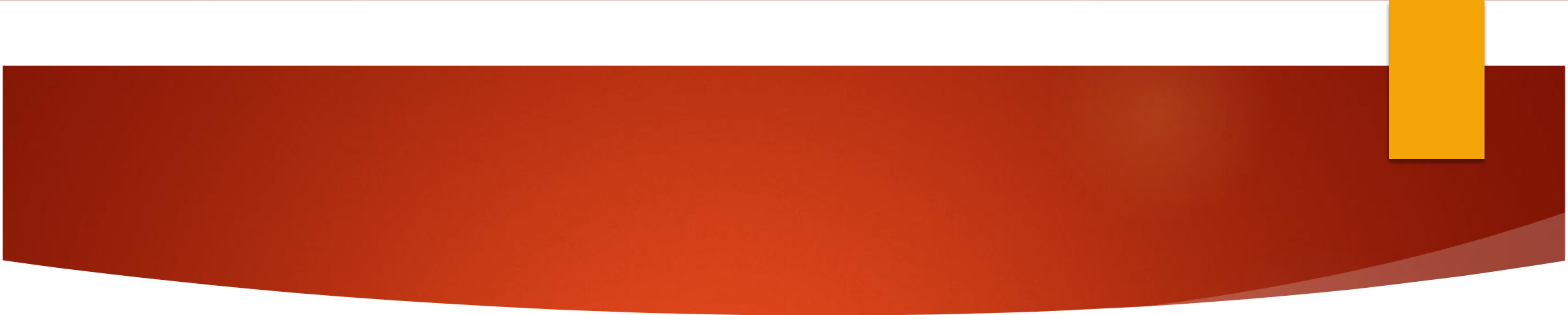
(+2 SENT-LIM); / * *and following which is a sentence boundary,* * /

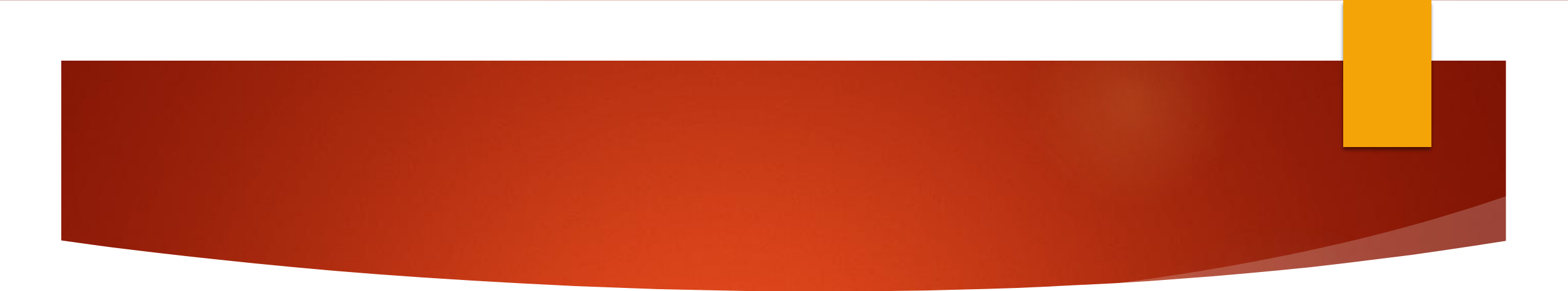
(NOT -1 SVOC/A); / * *and the previous word is not a verb like* * /

/ * *'consider' which allows adjs as object complements* * /

then eliminate non-ADV tags

else eliminate ADV tag

- 
- ▶ The **first two clauses** of this rule check to see that the that directly precedes a sentence-final adjective, adverb, or quantifier.
 - ▶ In all other cases the adverb reading is eliminated.
 - ▶ The **last clause** eliminates cases preceded by verbs like consider or believe which can take a noun and an adjective.



Tutorial

Problem No: 1

Find one tagging error in each of the following sentences that are tagged with the Penn Treebank tagset:

- a. I/PRP need/VBP a/DT flight/NN from/IN Atlanta/NN
- b. Does/VBZ this/DT flight/NN serve/VB dinner/NNS
- c. I/PRP have/VB a/DT friend/NN living/VBG in/IN Denver/NNP
- d. What/WDT flights/NNS do/VBP you/PRP have/VB from/IN Milwaukee/NNP
to/IN Tampa/NNP
- e. Can/VBP you/PRP list/VB the/DT nonstop/JJ afternoon/NN flights/NNS

Problem No: 2

Use the Penn Treebank tagset to tag each word in the following sentences from Damon Runyon's short stories. You may ignore punctuation.

- a. It is a nice night.
- b. This crap game is over a garage in Fifty-second Street. . .
- c. . . . Nobody ever takes the newspapers she sells . . .
- d. He is a tall, skinny guy with a long, sad, mean-looking kisser, and a mournful voice.

STOCHASTIC PART-OF-SPEECH TAGGING

- ▶ Statistical (or stochastic) taggers use a training corpus to compute the probability of a tag in a context.
- ▶ For a given word sequence, Hidden Markov Model (HMM) Taggers choose the tag sequence that maximizes

$$P(\text{word} \mid \text{tag}) * P(\text{tag} \mid \text{previous-n-tags})$$

A bigram HMM tagger chooses the tag t_i for word w_i that is most probable given the previous tag, t_{i-1}

$$t_i = \operatorname{argmax}_j P(t_j \mid t_{i-1}, w_i)$$

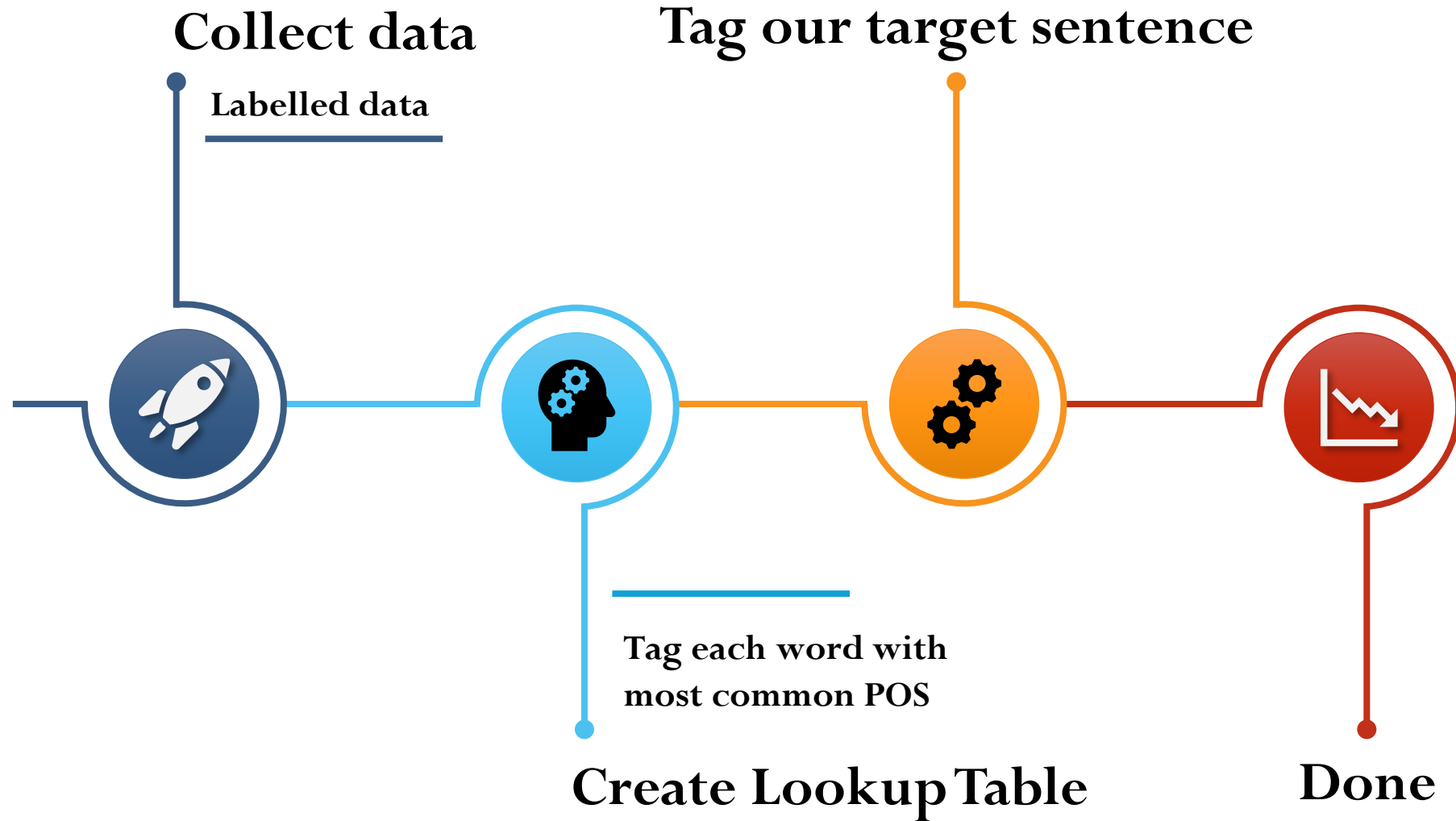
POS tagging with Hidden Markov Model



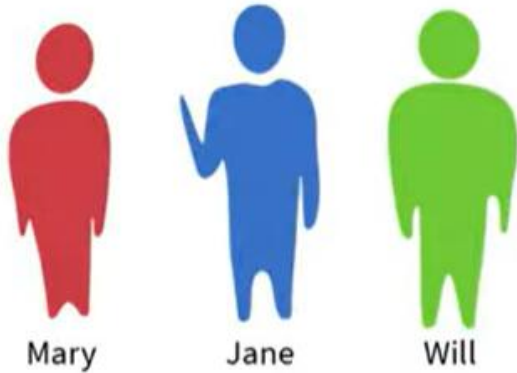
Jane	Saw	Will
Noun	Verb	Noun

Mary	Saw	Will
Noun	Verb	Noun

Lookup Table



Contd...



Jane	Saw	Will
Noun	Verb	Noun

Mary	Saw	Will
Noun	Verb	Noun

	Noun	Verb
Mary	1	0
Saw	0	2
Jane	1	0
will	2	0

Jane	will	See	Will
Noun	modal	Verb	Noun
Mary	will	See	Jane
Noun	modal	Verb	Noun
will	will	See	Mary
Noun	modal	Verb	Noun





Mary will see Will

 ? ? ? ?

Jane	will	See	Will
Noun	modal	Verb	Noun
Mary	will	See	Jane
Noun	modal	Verb	Noun
will	will	See	Mary
Noun	modal	Verb	Noun

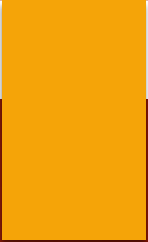
	Noun	Verb	Modal
Mary	2	0	0
See	0	3	0
Jane	2	0	0
will	2	0	3



Mary	will	see	Will
Noun	Modal	verb	?

Jane	will	See	Will
Noun	modal	Verb	Noun
Mary	will	See	Jane
Noun	modal	Verb	Noun
will	will	See	Mary
Noun	modal	Verb	Noun

	Noun	Verb	Modal
Mary	2	0	0
See	0	3	0
Jane	2	0	0
will	2	0	3



Contd...

Look up table wont work if a single word have multiple tag?

Wont consider any context



Mary will see Will
Noun Modal verb Noun

Jane will See Will

Noun modal Verb Noun

Mary will See Jane

Noun modal Verb Noun

will will See Mary

Noun modal Verb Noun

BIGRAM

	N-M	M-V	V-N
Mary-will	1	0	0
Will-See	0	3	0
See-Jane	0	0	1
Will-will	1	0	0
See-mary	0	0	1
Jane-will	1	0	0
See-will	0	0	1

Another Example

Our data

Mary Jane can see Will

Spot will see Mary

Will Jane spot Marry?

Marry will pat Spot

Jane	will	spot	will
?	?	?	?



Mary



Jane



Will



Spot

Our data

Mary Jane can see Will

Spot will see Mary

Will Jane spot Marry?

Marry will pat Spot

Jane will spot will
? ? ? ?

	N-M	M-V	V-N	etc...
mary-jane				
jane-can				
can-see				
see-will				
spot-will				
will-see				
see-mary				
will-jane				
jane-spot				
spot-mary				
mary-will				
will-pat				
pat-spot				

Jane will spot will
noun modal verb noun

We will need two things

Two probabilities

Emission probabilities

How likely is that Jane will be a noun, will be a modal,...

Transition Probabilities

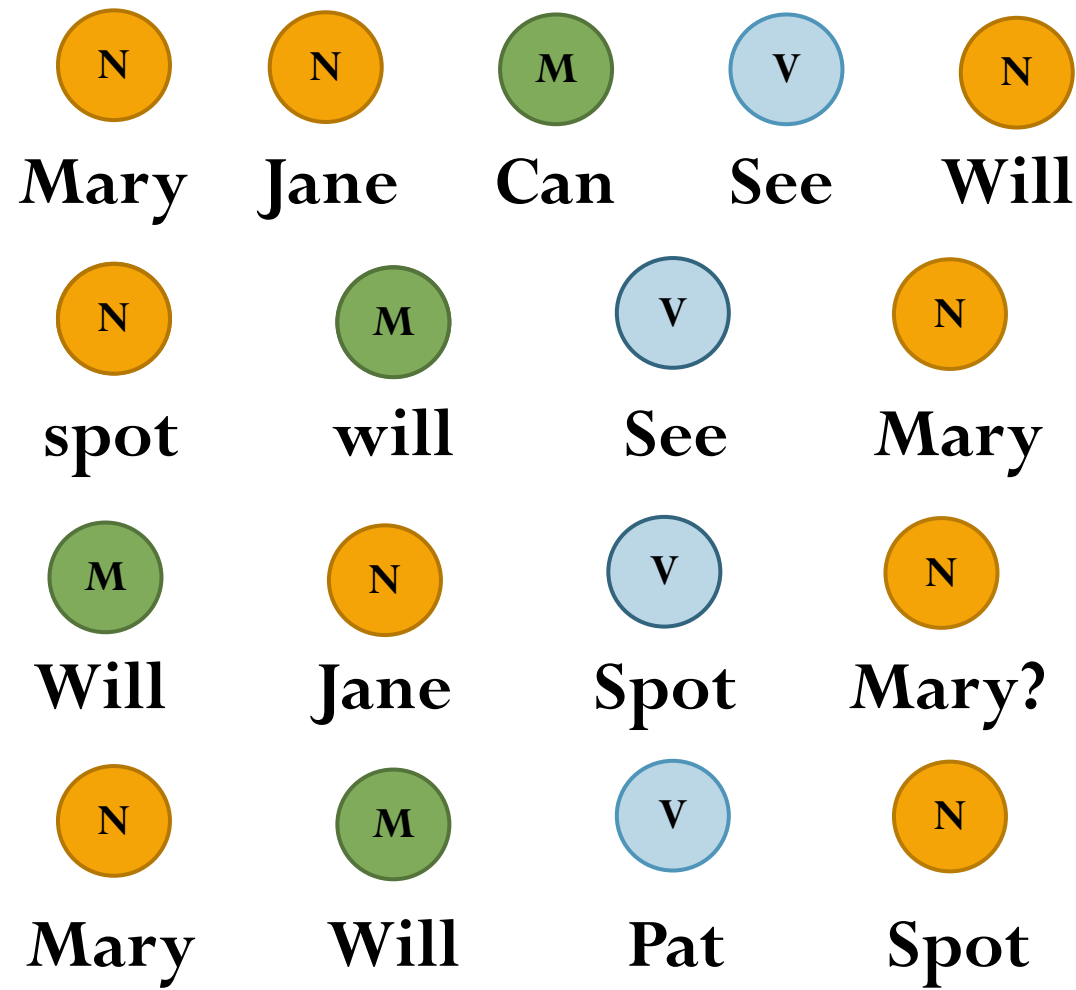
How likely is it that...

Noun is followed by a modal
which is followed by a verb and
than a noun



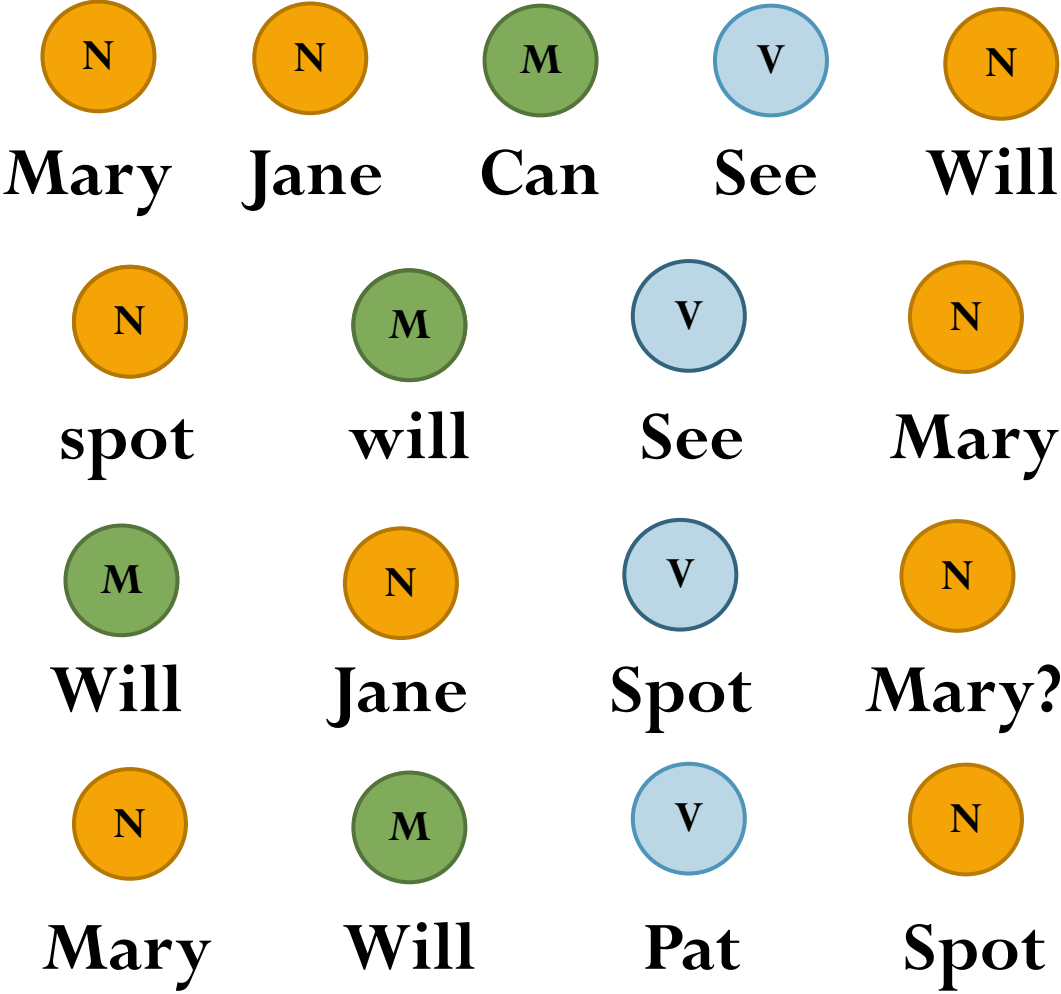
Emission Probabilities

	N	M	V
Mary	4	0	0
Jane	2	0	0
will	1	3	0
Spot	2	0	1
Can	0	1	0
See	0	0	2
pat	0	0	1



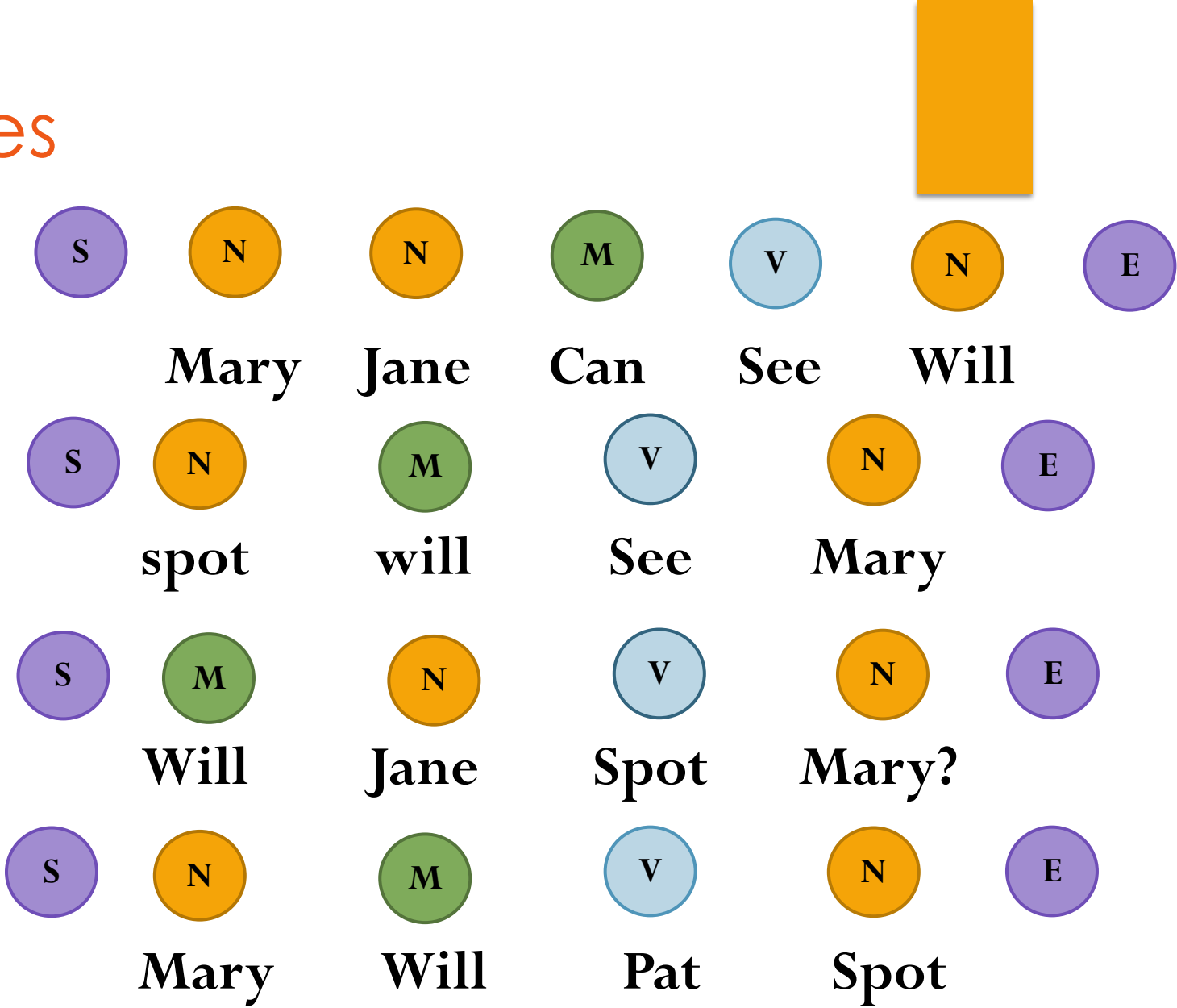
Emission Probabilities

	N	M	V
Mary	4/9	0	0
Jane	2/9	0	0
will	1/9	3/4	0
Spot	2/9	0	1/4
Can	0	1/4	0
See	0	0	2/4
pat	0	0	1/4

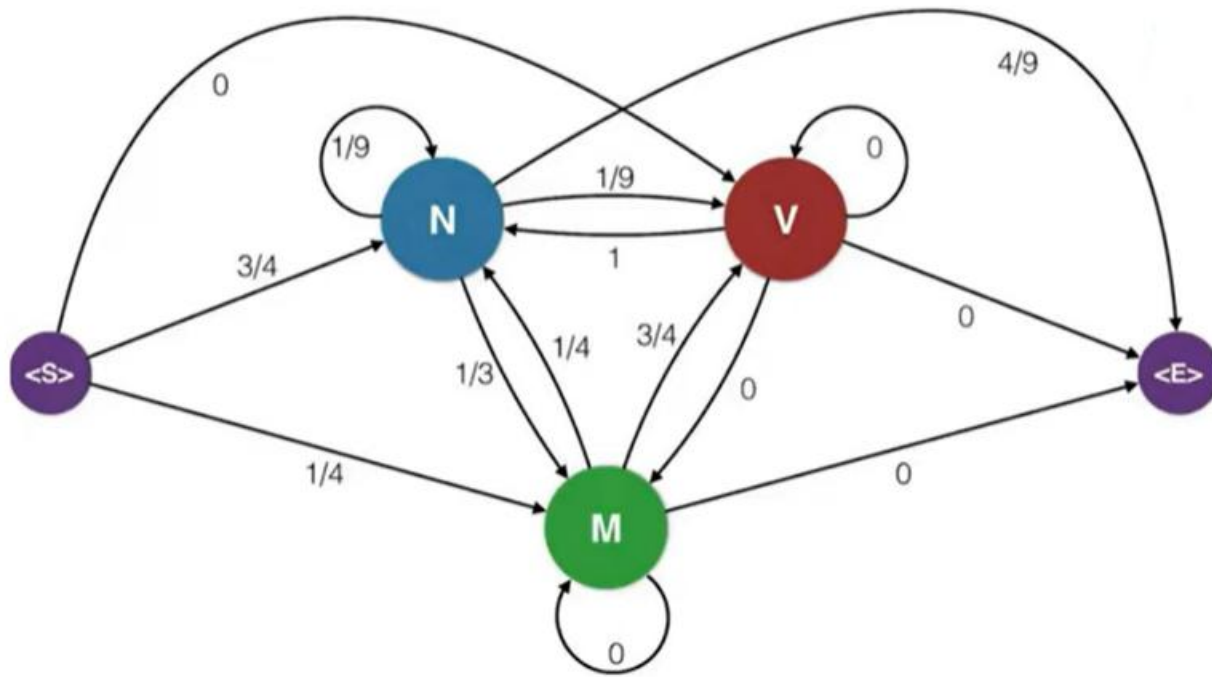


Transition Probabilities

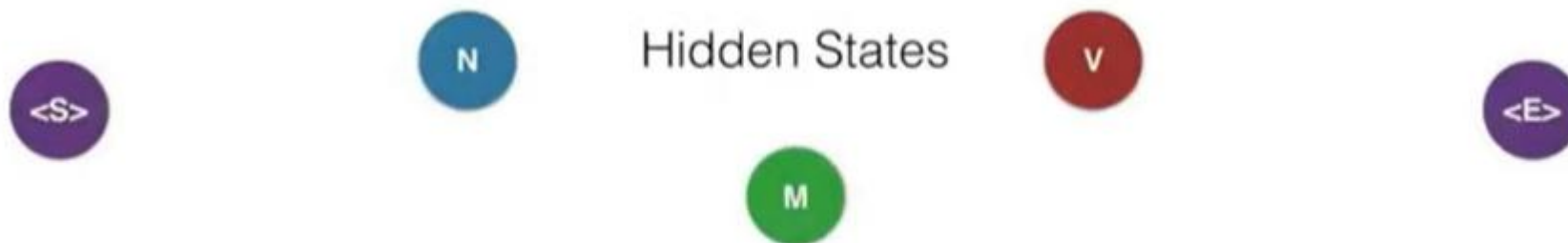
	N	M	V	<E>
<S>	3	1	0	0
N	1	3	1	4
M	1	0	3	0
V	4	0	0	0



Transition Probabilities



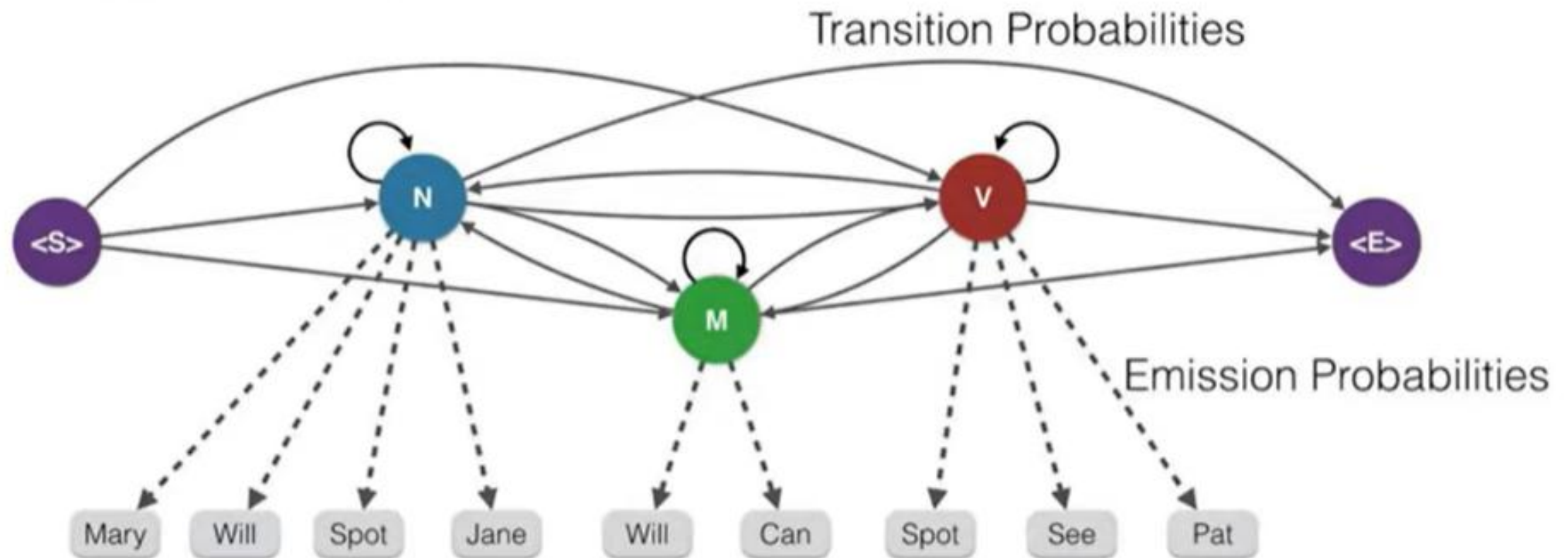
	N	M	V	<E>
<S>	3/4	1/4	0	0
N	1/9	3/9	1/9	4/9
M	1/4	0	3/4	0
V	4/4	0	0	0

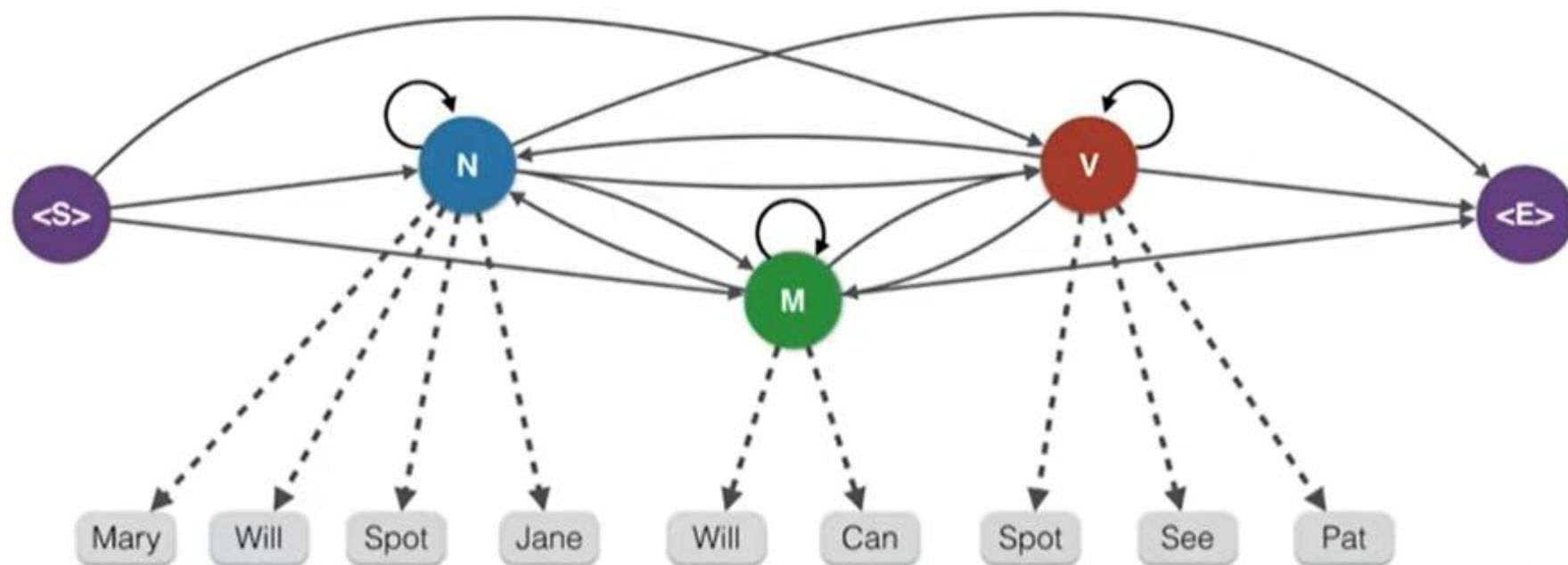


Mary Will Spot Jane Will Can Spot See Pat

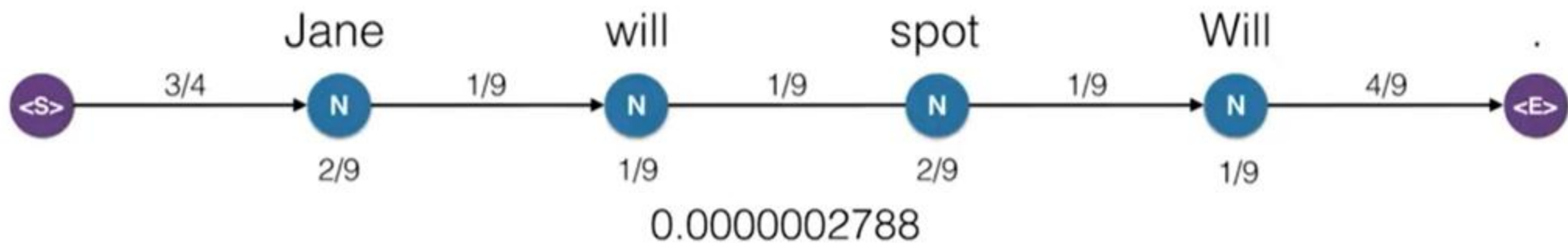
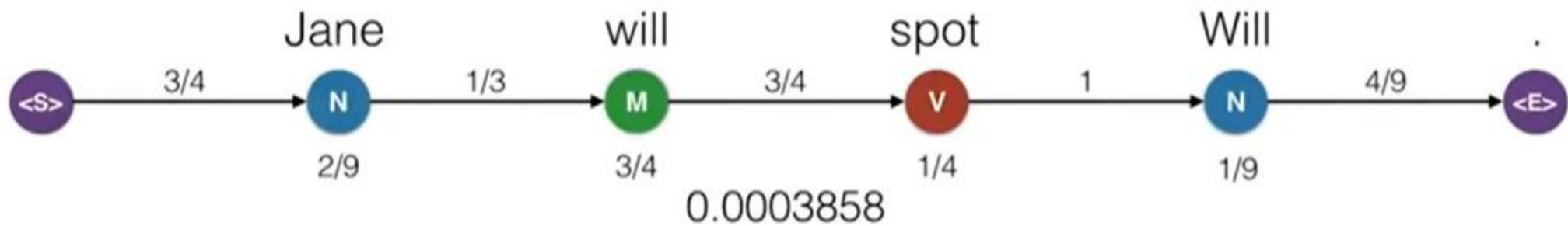
Observations

Hidden Markov Model



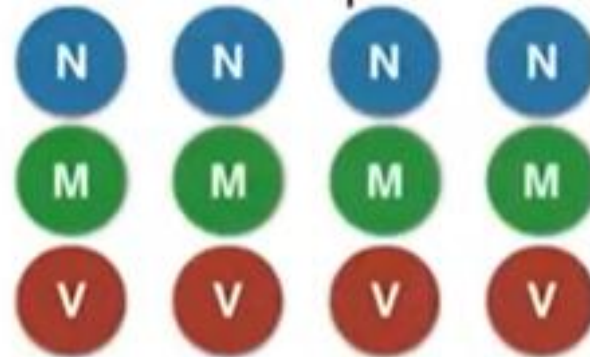


0.0003858



81 possibilities

Jane will spot Will.



Simple Example

From the Brown Corpus

- ▶ Secretariat/NNP is/VBZ expected/VBN to/TO *race*/VB tomorrow/NN
- ▶ People/NNS continue/VBP to/TO inquire/VB the/DT reason/NN
for/IN the/DT *race*/NN for/IN outer/JJ space/NN

Assume previous words have been tagged, and we want to tag the word *race*.

Bigram tagger

- ▶ to/TO *race*/?
- ▶ the/DT *race*/?

Example

- ▶ Goal: choose between NN and VB for the sequence *to race*
- ▶ Plug these into our bigram HMM tagging equation:

$$P(\text{word} \mid \text{tag}) * P(\text{tag} \mid \text{previous-n-tags})$$

- ▶ $P(\text{race} \mid \text{VB}) * P(\text{VB} \mid \text{TO})$
- ▶ $P(\text{race} \mid \text{NN}) * P(\text{NN} \mid \text{DT})$

How do we compute the tag sequence probabilities and the word likelihoods?

Word Likelihood

- ▶ We must compute the likelihood of the word *race* given each tag.
I.e., $P(\text{race} \mid \text{VB})$ and $P(\text{race} \mid \text{NN})$
- ▶ Note: we are **NOT** asking which is the most likely tag for the word.
- ▶ Instead, we are asking, if we were expecting a verb, how likely is it that this verb would be *race*?
- ▶ From the Brown and Switchboard Corpora:
 $P(\text{race} \mid \text{VB}) = .00003$
 $P(\text{race} \mid \text{NN}) = .00041$

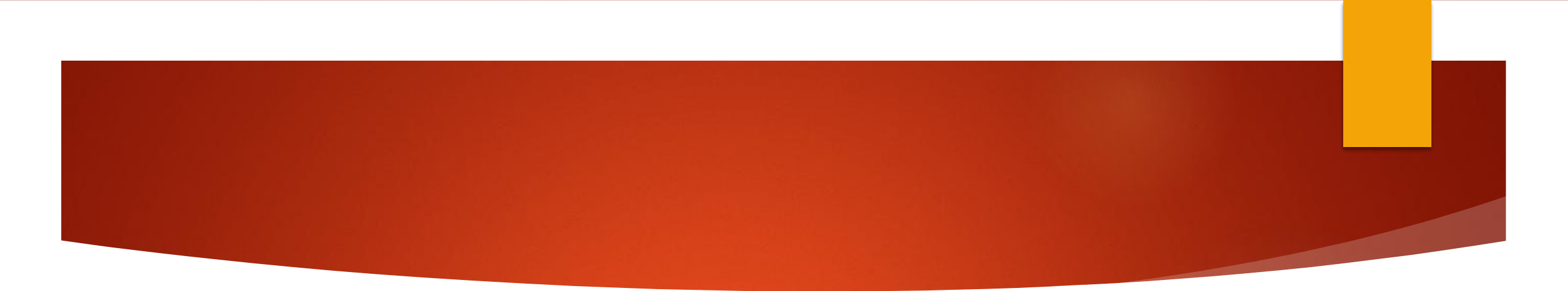
Tag Sequence Probabilities

- ▶ Computed from the corpus by counting and normalizing.
- ▶ We expect VB more likely to follow TO because infinitives (*to race, to eat*) are common in English, but it is possible for NN to follow TO (*walk to school, related to fishing*).

- ▶ From the Brown and Switchboard corpora:

$$P(\text{VB} \mid \text{TO}) = .340$$

$$P(\text{NN} \mid \text{TO}) = .021$$



Multiplying tag sequence probabilities by word likelihoods
gives

▶ $P(\textit{race} \mid \text{VB}) * P(\text{VB} \mid \text{TO}) = .000010$

▶ $P(\textit{race} \mid \text{NN}) * P(\text{NN} \mid \text{TO}) = .0000007$

So, even a simple bigram version correctly tags *race* as a
VB, despite the fact that it is the less likely sense.

The Actual Algorithm for HMM tagging

- ▶ Goal: choose the best sequence of tags T for a sequence of words W in a sentence

$$T' = \underset{T \in \tau}{\operatorname{argmax}} P(T|W)$$

- ▶ By Bayes Rule (giving us something easier to calculate)

$$P(T|W) = \frac{P(T)P(W|T)}{P(W)}$$

- ▶ Since we can ignore $P(W)$, we have

$$T' = \underset{T \in \tau}{\operatorname{argmax}} P(T)P(W|T)$$

Statistical POS Tagging: the Prior

$$P(T) = P(t_1, t_2, \dots, t_{n-1}, t_n)$$

By the Chain Rule:

$$= P(t_n \mid t_1, \dots, t_{n-1}) P(t_1, \dots, t_{n-1})$$

$$\prod_{i=1}^n P(t_i \mid t_1^{i-1})$$

Making the Markov assumption:

$$\approx P(t_i \mid t_{i-N+1}^{i-1}) \quad \text{e.g., for bigrams, } \prod_{i=1}^n P(t_i \mid t_{i-1})$$

the (Lexical) Likelihood

$$P(W | T) = P(w_1, w_2, \dots, w_n | t_1, t_2, \dots, t_n)$$

From the Chain Rule:

$$= \prod_{i=1}^n P(w_i | w_1 t_1 \dots w_{i-1} t_{i-1} t_i)$$

Simplifying assumption: probability of a word depends only on its

$$\text{own tag } P(w_i | t_i) \approx \prod_{i=1}^n P(w_i | t_i)$$

$$\text{So... } T' = \underset{T \in \tau}{\operatorname{argmax}} \prod_{i=1}^n P(t_i | t_{i-1}) \prod_{i=1}^n P(w_i | t_i)$$

Estimate the Tag Priors and the Lexical Likelihoods from Corpus

- ▶ Maximum-Likelihood Estimation

- ▶ For bigrams:

$$P(t_i | t_{i-1}) = c(t_{i-1}, t_i) / c(t_{i-1})$$

$$P(w_i | t_i) = \frac{c(w_i t_i)}{c(t_i)}$$

Performance

- ▶ This method has achieved 95-96% correct with reasonably complex English tagsets and reasonable amounts of hand-tagged training data.

Transformation-Based (Brill) Tagging

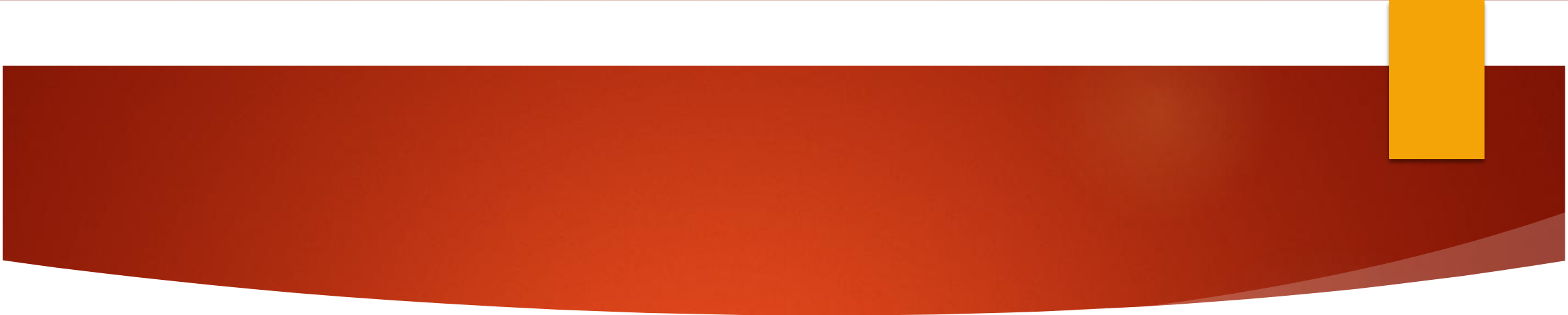
A hybrid approach

- ▶ Like rule-based taggers, this tagging is based on rules
- ▶ Like (most) stochastic taggers, rules are also automatically induced from hand-tagged data

Basic Idea: do a quick and dirty job first, and then use learned rules to patch things up

Overcomes the pure rule-based approach problems of being too expensive, too slow, too tedious etc...

An instance of **Transformation-Based Learning**.

- 
- ▶ Combine rules and statistics...
 - ▶ Start with a dumb statistical system and patch up the typical mistakes it makes.
 - ▶ How dumb?
 - ▶ Assign the most frequent tag (unigram) to each word in the input

Examples

- ▶ Race
 - ▶ “race” as NN: .98
 - ▶ “race” as VB: .02
- ▶ So you’ll be wrong 2% of the time, which really isn’t bad
- ▶ Patch the cases where you know it has to be a verb
 - ▶ Change NN to VB when previous tag is TO

Brill's Tagger 3 Stages

1. Label every word with its most likely tag.
2. Examine every possible transformation, and select the one that results in the most improved tagging.
3. Re-tag the data according to the selected rule.

Go to 2 until stopping criterion is reached.

Stopping: Insufficient improvement over previous pass.

Output: Ordered list of transformations. These constitute a tagging procedure.

Rule

- ▶ Where did that transformational rule come from?
- ▶ In principle, the set of possible rules is infinite.
 - ▶ Use set of rule templates to define possible rules to try in the search.

Hypothesis Space

- ▶ In Brill tagging it's defined by a set of templates of the form
 - ▶ Change tag a to tag b when ...
- ▶ a, b, w and z range over the tags

The preceding (following) word is tagged **z**.

The word two before (after) is tagged **z**.

One of the two preceding (following) words is tagged **z**.

One of the three preceding (following) words is tagged **z**.

The preceding word is tagged **z** and the following word is tagged **w**.

The preceding (following) word is tagged **z** and the word
two before (after) is tagged **w**.